

Parallel Computing with MPI

×

Today's goals

Complete any 3 learning items - 0/3

Complete a practice item

Complete 2 readings - 0/2

Module 1

Introduction to Parallel Computing and MPI

^

Welcome to the Course

Course Updates and Accessibility Support

Reading • 1 min

✓

Earn Academic Credit for your Work!

Reading • 10 min

✓

Course Support

Reading • 10 min

Course Overview

Video • 2 min

Foundations of Parallel Computing

Introduction to Parallel Computing

Video • 6 min

Thinking in Parallel

Video • 5 min

Message Passing and Synchronization

Introduction to Message Passing

Video • 4 min

Synchronization

Video • 3 min

Module Assignments

Module Quiz

Graded Assignment • 15 min

Distributed Weather Prediction System Using MPI

Programming Assignment • 1h

Module 2

Advanced Communication Techniques in MPI

▼

Module 3

Performance Optimization in Parallel Computing

▼

Module 4

Advanced MPI Concepts – Communicators and Derived Datatypes

▼

Module 5

Parallel I/O in MPI and HDF5 for High-Performance Computing

▼

Course Support

⏮

0:06 / 1:35

2x

🔊

⚙️

Non-Credit Learners

Instructional Support

Similar to other courses on Coursera, the non-credit experience is self-paced and asynchronous. If you have any questions or need help with assignments or course material, you can seek assistance from your fellow learners in the discussion forums or do some research on your own outside of the course. Please note that course facilitators can only provide instructional support and help with grading questions to learners in the for-credit version of the course.

Technical Support

For technical or platform-related support we recommend the following resources in the order listed:

- Review articles in Coursera's [Learner Help Center](#)
- Contact Coursera's [support team](#)
- [Flag non-urgent issues](#) (e.g. broken links, minor assignment errors)
- Post a message in the [Course Support Forum](#) if you encounter urgent issues that impede your learning progress. Please note that course facilitators follow the academic calendar, so their responses may be delayed during breaks, holidays, or when for-credit courses are not in session. Please refer to the [CU Degrees on Coursera Calendar](#) for session schedules.

For-Credit Learners

In addition to the resources listed above, for-credit learners have full access to course facilitators who:

- Monitor all discussion posts in the for-credit courses
- Offer regular office hours, where they can answer questions about content and provide assignment support
- Can be contacted directly through the course support email provided in the *Course Facilitator* reading

Go to next item

✓ Completed

👍 Like

👎 Dislike

📩 Report an issue